next chapter)—embody time-reversal symmetry: motion that can occur in the usual forward-time direction can equally well occur in reverse. As the terminology can be a bit confusing, let me reemphasize that we are not reversing time. Time is doing what it always does. Instead, our conclusion is that *we can make an object trace its trajectory in reverse by the simple procedure of reversing its velocity at any point along its path.* Equivalently, the same procedure—reversing the object's velocity at some point along its path— would make the object execute the motion we'd see in a reverse-run film.

Tennis Balls and Splattering Eggs

Watching a tennis ball shoot between Venus and Jupiter—in either direction—is not particularly interesting. But as the conclusion we've reached is widely applicable, let's now go someplace more exciting: your kitchen. Place an egg on your kitchen counter, roll it toward the edge, and let it fall to the ground and splatter. To be sure, there is a lot of motion in this sequence of events. The egg falls. The shell cracks apart. Yolk splatters this way and that. The floorboards vibrate. Eddies form in the surrounding air. Friction generates heat, causing the atoms and molecules of the egg, floor, and air to jitter a little more quickly. But just as the laws of physics show us how we can make the tennis ball trace its precise path in reverse, the same laws show how we can make every piece of eggshell, every drop of yolk, every section of flooring, and every pocket of air exactly trace its motion in reverse, too. "All" we need do is reverse the velocity of each and every constituent of the splatter. More precisely, the reasoning used with the tennis ball implies that if, hypothetically, we were able to simultaneously reverse the velocity of *every* atom and molecule involved directly or indirectly with the splattering egg, *all* the splattering motion would proceed in reverse.

Again, just as with the tennis ball, if we succeeded in reversing all these velocities, what we'd see would look like a reverse-run film. But, unlike the tennis ball's, the egg-splattering's reversal of motion would be extremely impressive. A wave of jostling air molecules and tiny floor vibrations would converge on the collision site from all parts of the kitchen, causing every bit of shell and drop of yolk to head back toward the impact location. Each ingredient would move with exactly the same